

Algorithms for Sampling a Quantum Microcanonical Ensemble of Harmonic Oscillators at Potential Minima and Conical Intersections

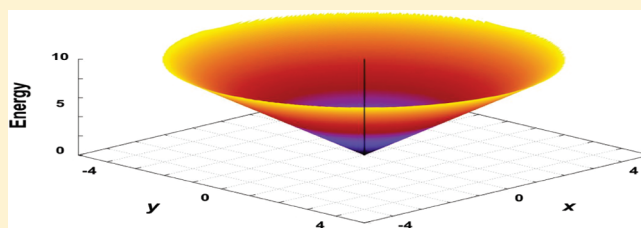
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ABSTRACT: Algorithms are presented for sampling quantum microcanonical ensembles for a potential energy minimum and for the conical intersection at the minimum energy crossing point of two coupled electronic states. These ensembles may be used to initialize trajectories for chemical dynamics simulations. The unimolecular dynamics of a microcanonical ensemble about a potential energy minimum may be compared with the dynamics predicted by quantum Rice–Ramsperger–Kassel–Marcus (RRKM) theory. If the dynamics is non-RRKM, it will be of particular interest to determine which states have particularly long lifetimes. Initializing a microcanonical ensemble for the electronically excited state at a conical intersection is a model for electronic nonadiabatic dynamics. The trajectory surface-hopping approach may be used to study the ensuing chemical dynamics. A strength of the model is that zero-point energy conditions are included for the initial nonadiabatic dynamics at the conical intersection.



I. INTRODUCTION

Classical trajectory chemical dynamics simulations are widely used^{1,2} to study a wide variety of chemical processes including unimolecular³ and bimolecular⁴ reactions, intramolecular vibrational energy redistribution (IVR),^{5,6} gas-phase collisional energy transfer,⁷ and collisions of projectiles with surfaces.^{8,9} To compare with experiment or a theoretical model for chemical dynamics, it is important to choose proper initial conditions (i.e., coordinates and momenta)¹⁰ for the ensemble of trajectories propagated for the simulation. The appropriate distribution functions must be sampled to model the chemical system and process under investigation.

In previous work an algorithm was described for Monte Carlo sampling of a microcanonical ensemble of classical harmonic oscillators.¹¹ Because of the energy continuity of classical mechanics, it was possible to derive an analytic expression for sampling this ensemble. This algorithm is important for selecting trajectories to test for intrinsic non-RRKM (Rice–Ramsperger–Kassel–Marcus) behavior in classical unimolecular dissociation^{3,12} and for modeling experiments for which the dynamics is well-represented by classical mechanics.¹³ However, for many situations, the classical description of the initial states is not correct, and the quantum representation is necessary.¹⁰ In this work an algorithm is presented for sampling a quantum microcanonical ensemble of harmonic oscillators about a potential energy minimum. Also presented is an algorithm for sampling a microcanonical ensemble for a conical intersection (CoI).^{14,15} This latter work was in part motivated by the manner in which Burghardt and co-workers^{16,17} partitioned modes in their studies of nonadiabatic dynamics.

Methods for sampling a microcanonical ensemble for a CoI, different than the algorithm presented here, have been used in previous chemical dynamics simulations.^{18–21} Lischka and co-workers¹⁸ have used two models in which the kinetic energy of the atoms is added randomly at the CoI, and the chemical dynamics is then propagated on the ground state potential energy surface (PES). For one model, the kinetic energy was only added to the two degrees of freedom for the *g,h* branching space defining the CoI, while for the other the kinetic energy was added to all the degrees of freedom. Bowman and co-workers^{19–21} performed an appropriate microcanonical sampling of momenta at the CoI for two different models of quenching OH by H₂, followed by ground state trajectories.

The CoI microcanonical sampling algorithm presented here is different than those described above. The sampling is not performed at the CoI and, instead, is performed about the CoI for the excited electronic state. It is meant to generate initial conditions for the nonadiabatic dynamics, which is not necessarily very fast when the starting point is substantially higher in energy than the CoI (the two limiting cases of slow and fast decay through a CoI correspond to the “adiabatic” and “diabatic” models of Bowman and co-workers).^{19–21} The remainder of this article is organized as follows: Section II describes sampling a microcanonical ensemble for a collection of normal mode harmonic oscillators. Section III describes the more complex algorithm that samples a microcanonical ensemble for a CoI. Section IV is a summary.

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II. QUANTUM MICROCANONICAL ENSEMBLE OF HARMONIC OSCILLATORS

Selecting initial coordinates and momenta for an ensemble of normal mode harmonic oscillators, with a fixed number of quanta in each mode, is a known procedure.¹⁰ What is required is the transformation of the quantum numbers $\mathbf{n}$ for this state to the coordinates and momenta (or velocities) required to calculate the classical trajectory. This can be done by choosing a different random classical phase for each normal mode¹⁰ or using a quantum mechanical distribution that simultaneously samples both the coordinates and momenta.^{22–24}

What has not been well-described is a procedure for sampling a quantum microcanonical ensemble of normal mode harmonic oscillator states at a potential energy minimum. An algorithm for this sampling procedure is described below. It is then used as a component for sampling a microcanonical ensemble for a CoI.

A. Algorithm. Consider a molecule consisting of s quantum harmonic oscillators at a total constant energy

$$E = \sum_{i=1}^s E_i \quad (1)$$

where the $E_i = n_i h\nu_i$ ($n_i = 0, 1, 2, \dots$) are the individual harmonic oscillator energies. For a microcanonical ensemble, the probability that oscillator 1 has energy E_1 is proportional to the density of states of the remaining $s - 1$ oscillators, which contain energy $E - E_1$; i.e., the density $\rho_{s-1}(E - E_1)$.¹¹ The most probable value of E_1 is zero, for which the density has its maximum value $\rho_{s-1}(E)$. A value for E_1 is chosen, as described below, by sampling the relative probability $\rho_{s-1}(E - E_1)/\rho_{s-1}(E)$. Once E_1 is chosen, E_2 is chosen randomly between 0 and $E - E_1$ in accord with the relative probability $\rho_{s-2}(E - E_1 - E_2)/\rho_{s-2}(E - E_1)$. To choose energy E_j for oscillator j , the relative probability

$$\rho_{s-j}(E - \sum_{i=1}^j E_i) / \rho_{s-j}(E - \sum_{i=1}^{j-1} E_i) \quad (2)$$

is sampled, where E_j is in the range 0 to $E - \sum_{i=1}^{j-1} E_i$.

For quantum harmonic oscillators, the quantum numbers n_i must be chosen for the oscillators, and this may be done by using the von Neumann rejection technique²⁶ to sample the relative probabilities of the quantum energy levels. This is illustrated here for choosing E_1 . To begin, a random value for E_1 is chosen by multiplying E by a freshly generated random number ranging between 0 and 1. A value for the quantum number n_1 is then chosen by “binning” the energy;² i.e., if E_1 is between 0 and $h\nu_1$, n_1 is set to zero; if it is between $h\nu_1$ and $2h\nu_1$, n_1 is set to one, and so forth. The energy for oscillator 1 is then set to $E_1 = n_1 h\nu_1$, and the value of E_1 is accepted if the relative probability $\rho_{s-1}(E - E_1)/\rho_{s-1}(E)$ is greater than a second random number. If not, a new trial value for E_1 is selected. The Beyer–Swinehart algorithm²⁷ is used to calculate the density of states ρ .

To select E_2 , the sampling range for the energy is reduced to $E - E_1$. The same procedure is then followed for the remaining oscillators, except for the last two oscillators s and $s - 1$. These two oscillators contain a total energy

$$E_{s,s-1} = E - \sum_{i=1}^{s-2} E_i \quad (3)$$

and, to identify their quantum numbers n_s and n_{s-1} , the total energy is assumed to have a resolution of $E \pm \Delta E$. All n_s and n_{s-1}

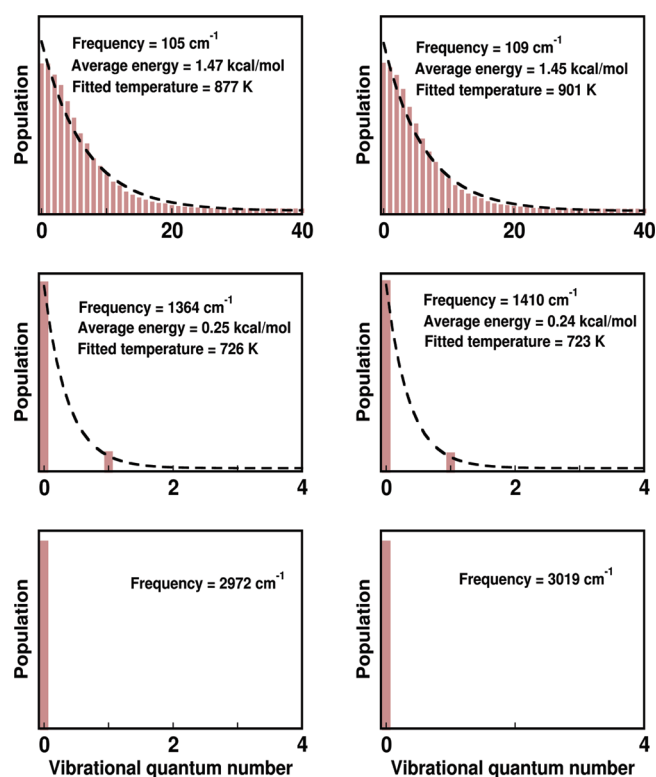


Figure 1. Distributions of quantum numbers for acetone, with total energy of 10 kcal/mol. Distributions are plotted for the normal modes with the two highest, two lowest, and two intermediate frequencies.

quantum states are identified, which have a total energy between $E_{s,s-1} \pm \Delta E$. One of these states is chosen randomly, with $E_{s,s-1} = E_s + E_{s-1}$.

To calculate a trajectory, these n_i quantum numbers for the s harmonic oscillator normal modes must be transformed to Cartesian momenta and coordinates. The standard way to do this is to choose a different random classical phase for each normal mode.^{10,28} To sample the coordinates and momenta for the $\mathbf{n} = 0$ state quantum mechanically, the Wigner distribution²² may be used. For states with $n > 0$, use of the Wigner distribution is problematic, since the Wigner function has negative regions, and the Husimi function^{23,24} may be used to obtain quantum probabilities for the coordinates and momenta of these states. The difference in the dynamics resulting from classical and quantum sampling of the coordinates and momenta is expected to be greatest for the $\mathbf{n} = 0$ state, since the difference between the classical and quantum distributions for the coordinates and momenta is greatest for this state.¹⁰ Thus, it is noteworthy that in recent work²⁵ it was found that sampling initial conditions at a transition state for the $\mathbf{n} = 0$ quantum state, by using either the classical random phase or the Wigner distribution, gave statistically the same trajectory results.

B. Application. The above model was applied to acetone for which the normal mode vibrational frequencies are 3019(2), 2972, 2963, 2937(2), 1731, 1454, 1435, 1426, 1410, 1364(2), 1216, 1091, 1066, 891, 877, 777, 530, 484, 385, 109, and 105.²⁹ Quantum states with quantum numbers $\mathbf{n}$ were selected randomly as described above for energies of 10, 50, and 100 kcal/mol with a resolution of $\Delta E = 100 \text{ cm}^{-1} = 0.3 \text{ kcal/mol}$. Sampling was performed with both 100 000 and 1 000 000 states selected at each energy, and the sampling with 100 000

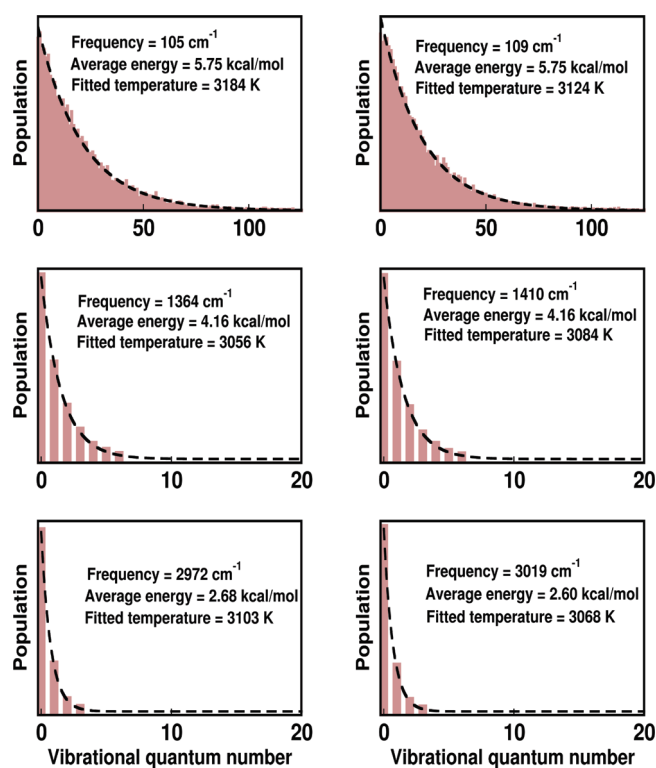


Figure 2. Same as Figure 1, except the total energy is 100 kcal/mol.

states is sufficient to illustrate the sampling statistics. For the sampling, the two lowest frequency modes, i.e., 109 and 105 cm^{-1} , were identified as modes s and $s - 1$. This is not necessary, but it makes the sampling more efficient. Identifying the two lowest frequency modes as s and $s - 1$ makes it more probable that there is a quantum state in the energy interval $E_{s,s-1} \pm \Delta E$ (i.e., see eq 3).

Distributions $P(n_i)$, of populating the n_i quantum levels for oscillator i , are plotted in Figures 1 and 2 for the two highest and two lowest frequency oscillators and for two oscillators with intermediate frequencies of 1410 and 1364 cm^{-1} . The distributions are for the total energies of 10 and 100 kcal/mol in Figures 1 and 2, respectively. At high energies it is well-known that each $P(n_i)$ distribution is well represented by the Boltzmann distribution^{30,31}

$$P(n_i) \propto e^{-n_i h \nu_i / k_B T_i} \quad (4)$$

and the fits to this distribution are also shown in Figures 1 and 2. It is of interest that for the low energy 10 kcal/mol sampling, the $P(n_i)$ distributions are well fit by the Boltzmann distribution. For this energy there is no population of the $n = 1$ levels of the 2972 and 3010 cm^{-1} modes for the 100 000 samplings.

Each $P(n_i)$ distribution may be characterized by its average energy

$$\langle E_i \rangle = \sum_{n_i=0} n_i h \nu_i P(n_i) / \sum_{n_i=0} P(n_i) \quad (5)$$

and its fitted temperature T_i . The values of $\langle E_i \rangle$ and T_i for each of the 24 acetone oscillators are compared in Table 1 for the energies of 10, 50, and 100 kcal/mol. As expected, the average energy in a mode increases with decrease in the mode frequency as the oscillator becomes more “classical-like”. This increase

Table 1. Average Energies and Fitted Temperatures for Sampling Quantum Microcanonical Ensembles for Acetone

mode ν	average energy (kcal/mol) ^a		
	10 ^b	50	100
3019	0.0 ^c	0.84 (1828)	2.60 (3068)
3019	0.0	0.82 (1801)	2.56 (3021)
2972	0.0	0.89 (1845)	2.68 (3103)
2963	0.0	0.92 (1876)	2.64 (3105)
2937	0.0	0.86 (1806)	2.66 (3038)
2937	0.0	0.86 (1810)	2.67 (3052)
1731	0.14 (706)	1.74 (1914)	3.74 (3064)
1454	0.23 (736)	1.99 (1917)	4.07 (3148)
1435	0.24 (733)	2.01 (1934)	4.12 (3094)
1426	0.24 (732)	2.03 (1953)	4.20 (3204)
1410	0.24 (723)	2.03 (1932)	4.16 (3084)
1364	0.25 (726)	2.09 (1952)	4.16 (3056)
1364	0.25 (723)	2.08 (1902)	4.24 (3120)
1216	0.32 (732)	2.21 (1943)	4.40 (3132)
1091	0.41 (760)	2.26 (1861)	4.58 (3191)
1066	0.42 (759)	2.35 (1896)	4.56 (3177)
891	0.53 (771)	2.54 (1925)	4.79 (3191)
877	0.53 (768)	2.57 (1929)	4.76 (3061)
777	0.61 (769)	2.71 (1994)	4.90 (3099)
530	0.84 (799)	2.96 (1903)	5.13 (3100)
484	0.87 (788)	3.04 (1944)	5.38 (3165)
385	0.97 (797)	3.08 (1882)	5.51 (3122)
109	1.45 (901)	3.59 (2089)	5.75 (3124)
105	1.47 (878)	3.57 (2026)	5.75 (3184)

^a The values in the parentheses are the fitted temperatures (K) for the modes' $P(n)$ distribution. Zero point energies are not included. ^b The total energies above the zero-point level are 10, 50, and 100 kcal/mol.

^c The energy is zero within the statistics of the 100 000 samplings. No states were selected with $n = 1$ or higher for any of the vibrational modes with ν of 2937–3019 cm^{-1} .

becomes less-pronounced as the total energy is increased. For the high energy limit, each oscillator will have an average energy of RT_i , with T_i given by $E = sRT_i$. The fitted T_i tend to increase as the mode frequency decreases.

III. SAMPLING A MICROCANONICAL ENSEMBLE FOR A CONICAL INTERSECTION

The model presented here assumes that the lifetime of the excited electronic state is sufficiently long, and the intramolecular dynamics of the excited electronic state's PES is of the form that a microcanonical ensemble is formed at the minimum energy crossing point of the excited electronic state PES with the PES of a lower energy electronic state. If there is a CoI^{14,15} at this crossing point, the algorithm described here may be used to prepare a microcanonical ensemble of initial conditions for the electronically excited state about this CoI. The electronic non-adiabatic dynamics for this ensemble may then be studied by a chemical dynamics simulation with surface-hopping.^{32,33} Of particular interest is the ensuing dynamics on the PES of the lower electronic state following the electronic transition. It is recognized that this is a model for the electronic nonadiabatic dynamics and is somewhat akin to the assumption of a microcanonical

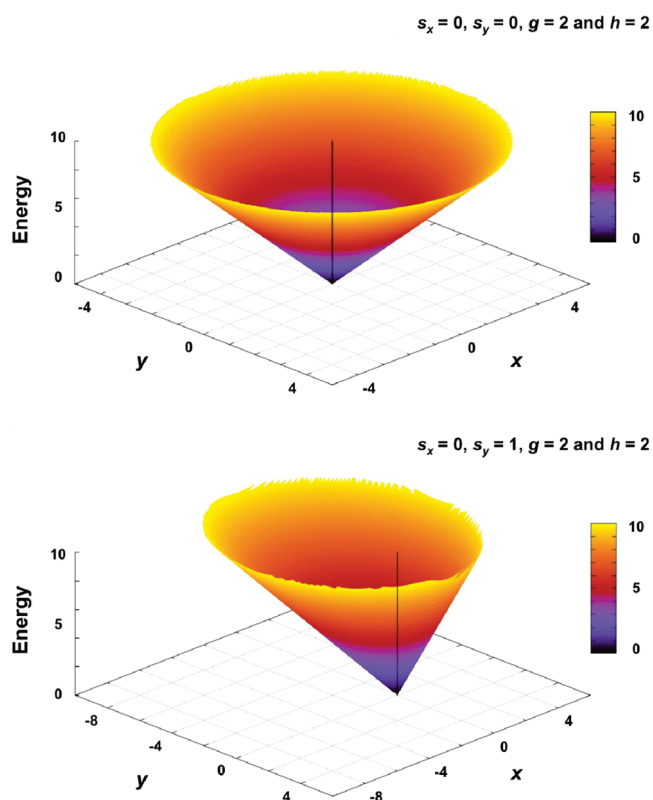


Figure 3. Examples of CoI's. The parameters for the vertical symmetric cone are $s_x = s_y = 0$ and $g = h = 2$. The parameters for the tilted symmetric cone are $s_x = 0$, $s_y = 1$, and $g = h = 2$. Units are in kcal/mol for the energy and angstroms for the x and y coordinates.

distribution of states at the transition state for a unimolecular decomposition reaction.^{34,35}

A. Hamiltonian. 1. *Mathematical Form.* If there is a CoI^{14,15} at the minimum energy crossing point of the PES for an electronically excited state with the PES of a lower electronic state, the intramolecular Hamiltonian for the excited state with s degrees of freedom is given by

$$H = H_{\text{cone}} + H_{s-2} \quad (6)$$

where H_{cone} is the Hamiltonian for the two-dimensional cone, and H_{s-2} is the Hamiltonian for the remaining $s - 2$ degrees of freedom.

The potential energy function for the cone is given by

$$V(x, y) = s_x x + s_y y + [(gx)^2 + (hy)^2]^{1/2} \quad (7)$$

and it has two components: one with s_x and s_y , and the other with g and h . Without s_x and s_y the potential is that for a perfect cone if $g = h$, and of an ellipsoid if $g \neq h$. The role of s_x and s_y is to tilt the cone, and s_x and s_y can be either positive or negative depending on which direction the cone should be tilted. To see the general form of the potential, consider the one-dimensional potential with $y = 0$. Then one has

$$V(x) = s_x x + g|x| \quad (8)$$

with the $g|x|$ term always positive; i.e., if $s_x = 0$ the $V(x)$ potential is symmetric. The role of s_x is to tilt the curve in either the $+x$ or $-x$ direction. If $g > s_x$, $V(x)$ will always be positive. For illustration, plots of two cones are shown in Figure 3.

One is for a perfect cone without tilt, and the other is a tilted cone.

In mass-weighted coordinates, the Hamiltonian for the cone is

$$H_{\text{cone}} = p_x^2/2 + p_y^2/2 + V(x, y) \quad (9)$$

For the potential energy $V(x, y)$, the unit for g , h , s_x and s_y is energy/(mass)^{1/2} · distance. The unit for p_x and p_y is (mass)^{1/2} · velocity.

For the model considered here, H_{s-2} is represented as a sum of Hamiltonians for $s - 2$ quantum harmonic oscillators. Thus, E_{s-2} is a sum of $n_i h \nu_i$ as discussed in the previous section. The molecule considered is acetone (described above). The C=O stretch, 1731 cm⁻¹, and C=O out-of-plane bend, 484 cm⁻¹, were removed as the two degrees of freedom for the cone.

2. *Transformation to Cartesian Coordinates and Momenta.* Unit vectors for the cone coordinates x and y are defined as¹⁵

$$\begin{aligned} \mathbf{x} &= \mathbf{g}/g \text{ and } \mathbf{y} = \mathbf{h}/h \\ \mathbf{g} &= \nabla(\Delta H) \text{ and } \mathbf{h} = \nabla(H_{12}) \end{aligned} \quad (10)$$

where ΔH is the difference in the electronic Hamiltonians for the two states, i.e., $(H_{22} - H_{11})/2$, and H_{12} is the coupling between the two states. The ∇ operator may be expressed in mass-weighted Cartesian coordinates, so that the transformation of x and y for the cone (as well as p_x and p_y) to Cartesian coordinates is given by the projections of $\mathbf{x}$ and $\mathbf{y}$ onto the Cartesian axes.

The eigenvalues (i.e., vibrational frequencies) and eigenvectors for the $s - 2$ normal modes, at the CoI, are found by diagonalizing a properly projected mass-weighted Cartesian force constant matrix. Projected from the matrix are the two degrees of freedom for the cone and the six degrees of freedom for translation and overall rotation. The eigenvector matrix provides the transformation from the momenta and coordinates for the $s - 2$ normal modes to the Cartesian momenta and coordinates.¹⁰

B. Algorithm. 1. *Choosing the Energy for the Cone, E_2 .* The total intramolecular energy for the excited electronic state is $E = E_2 + E_{s-2}$, where E_2 is the energy for the two-dimensional cone, and E_{s-2} is the energy for the remaining degrees of freedom. The former is treated classically, and the latter is treated as quantum harmonic oscillators.

The probability of a value for E_2 is proportional to $\rho(E, E_2) = \rho(E_2) \rho(E_{s-2})$, where $\rho(E_2)$ is the classical density of states for the cone's 2 degrees of freedom and $\rho(E_{s-2})$ is the quantum density of states for the $s - 2$ harmonic oscillators. The Beyer–Swinehart algorithm, as described above, may be used to find the density of states for the $s - 2$ quantum harmonic oscillators. The classical density of states for the cone is proportional to the derivative of the cone's phase space volume with respect to E_2 . Initial attempts were made to derive an analytic expression for the four-dimensional phase space volume of a general cone as given by eq 9, but these attempts were not successful. Thus, for the samplings reported here, Monte Carlo integration³⁶ was used to find the four-dimensional phase space volume. If one degree of freedom is removed from the cone, the resulting two-dimensional phase space volume, i.e., the x, p_x area, is straightforward to find and is proportional to $E_2^{3/2}$, suggesting the four-dimensional volume is proportional to E_2^3 . For all cone shapes, the volume was found to be proportional to E_2^3 . Thus, the classical density of states is proportional to E_2^2 . After the samplings were completed the

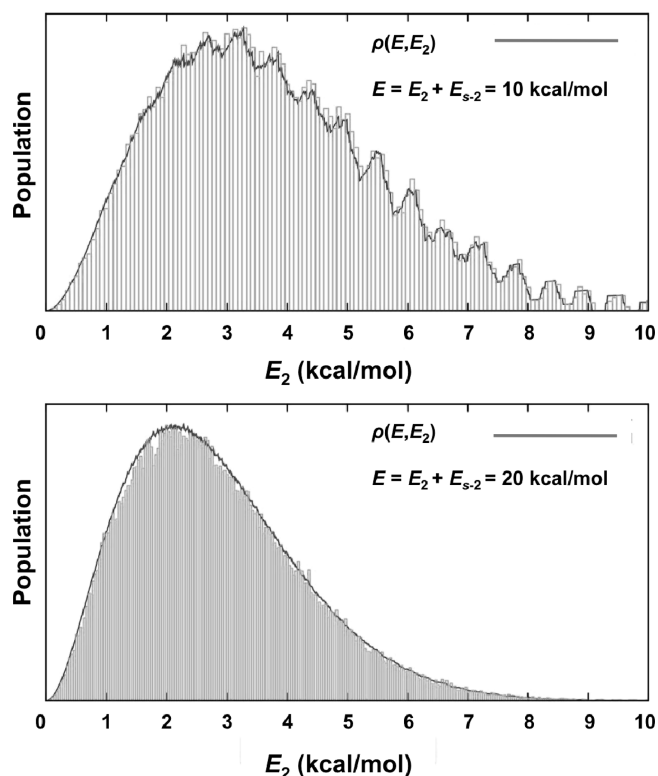


Figure 4. Distributions of energies for the cone's 2 degrees of freedom. Total energies of the molecule are 10 kcal/mol for the top plot, and 20 kcal/mol for the bottom plot.

analytic expression for the cone's phase space volume was derived and is presented in the Appendix.

The von Neumann rejection algorithm²⁶ is used to sample $\rho(E, E_2)$ and select a value for E_2 . The relative probability $\rho(E, E_2)/\rho^\circ(E, E_2)$ is sampled, where $\rho^\circ(E, E_2)$ is the most probable value for $\rho(E, E_2)$. Its value is determined by varying E_2 between 0 and E and finding the maximum in $\rho(E, E_2)$. Figure 4 gives plots of $\rho(E, E_2)$ for the acetone model with E equal to 10 and 20 kcal/mol, calculated using the expressions for $\rho(E_2)$ and $\rho(E_{s-2})$. Also given are histograms of $\rho(E, E_2)$, determined by sampling $\rho(E, E_2)$ with the von Neumann rejection technique. There are 1 000 000 samplings for each histogram. The agreement between the histograms and the actual $\rho(E, E_2)$ are quite good. At the higher E , $\rho(E, E_2)$ is nearly continuous. At the lower E , $\rho(E, E_2)$ has structure resulting from the discrete, noncontinuous nature of the quantum density of states $\rho(E_{s-2})$.

With E_2 chosen, the energy for the $s - 2$ normal modes is then $E_{s-2} = E - E_2$. E_{s-2} is distributed statistically between the normal modes and quantum numbers n_i are assigned as described above in Section II.A.

2. Determining the Initial Cartesian Coordinates and Momenta. To calculate the trajectory for the selected initial condition, the energy E_2 for the cone and the n_i quantum numbers for the $s - 2$ normal modes are transformed to Cartesian coordinates and momenta. The procedure for the latter transformation is discussed above in Section III.A.2.

The cone energy is treated classically, and x, y, p_x , and p_y for the cone are randomly selected so that a microcanonical ensemble is prepared at energy E_2 . A phase space point is chosen at random in the volume element $dp_x dp_y dx dy$ within E_2 to $E_2 + dE_2$. An

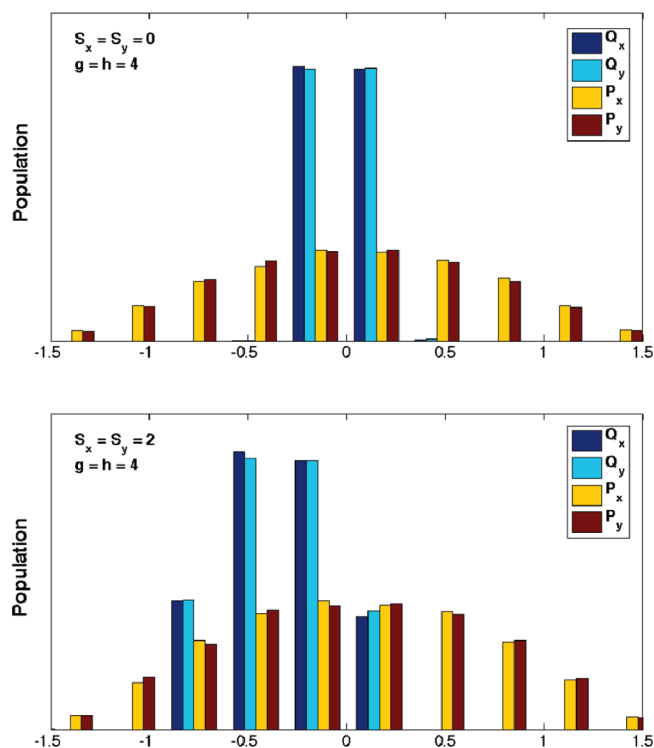


Figure 5. Distributions for the x, y, p_x , and p_y of the cone's Hamiltonian. The top plot is for a vertical symmetric cone, and the bottom plot is a tilted symmetric cone.

efficient method for doing this is to choose three coordinates and momenta randomly, and then find the remaining one by energy conservation.^{10,37} For the sampling used here, the remaining variable is p_y and the proper weighting for selecting p_y is seen from the transformation

$$dx dy dp_x dp_y = dx dy dp_x dH_2 |\partial H_2 / \partial p_y|^{-1} \quad (11)$$

where H_2 is the Hamiltonian for the cone. The relative weighting factor for choosing p_y is

$$P(p_y) \propto |\partial H_2 / \partial p_y|^{-1} \quad (12)$$

which is $|p_y|^{-1}$ for the cone's Hamiltonian. The von Neumann rejection method may be used to sample this relative probability.

Distribution of x, y, p_x , and p_y , randomly sampled for the cone in the above manner are plotted in Figure 5. Two types of cones were selected: vertical symmetric and tilted symmetric. The energy is $E_2 = 10$ kcal/mol, and the parameters for the cone's potential energy function $V(x, y)$ are $s_x = s_y = 0$ and $g = h = 4$ for the vertical symmetric cone, and $s_x = s_y = 2$ and $g = h = 4$ for the tilted symmetric cone (units are the integration units for the program VENUS³⁸). For the vertical symmetric cone, the distributions are peaked at zero, as expected, and the x and y distributions are statistically the same as are the p_x and p_y distributions, which is the required result given the form of the cone's Hamiltonian. In the case of the tilted cone, the peaks of the distributions for x and y are shifted toward the direction for which the cone is tilted.

IV. SUMMARY

In the work presented here, algorithms are described for selecting a microcanonical ensemble of quantum vibrational states at a potential energy minimum and for a CoI at the minimum energy crossing point between two coupled electronic states. The microcanonical sampling at a potential energy minimum may be used for chemical dynamics simulations of unimolecular decomposition. It will be of particular interest to compare the calculated unimolecular rate constant for the initial quantum microcanonical ensemble with the rate constant of quantum RRKM theory.¹² If the unimolecular dynamics of this ensemble is intrinsically non-RRKM,³ it will be of interest to determine which states in the ensemble have particularly long lifetimes with respect to unimolecular decomposition.

In previous work,^{18–21} algorithms have been presented for selecting a microcanonical (or random) distribution of states at a CoI. The model proposed here is related to, but is different than, this work in that the microcanonical ensemble is not prepared at the CoI but about the CoI. For the s internal degrees of freedom at the CoI, the two degrees of freedom for the cone are treated classically and the remaining $s - 2$ degrees of freedom are represented by quantum harmonic oscillator normal modes of vibration. The total energy is distributed between the cone and the $s - 2$ degrees of freedom in accord with a microcanonical ensemble. The quantum harmonic oscillator normal modes of vibration are sampled in the same manner as for the sampling at a potential energy minimum. The coordinates and momenta for the cone are selected by classical microcanonical sampling. The trajectory surface-hopping method,^{32,33} in which all couplings between the two electronic states are taken into account, may then be used to study the electronic nonadiabatic dynamics of this initial microcanonical ensemble. Previous comparisons^{39,40} between mixed classical/quantum simulations and fully quantum ones have shown that, except for very small total energies at the CoI, the surface-hopping and quantum nonadiabatic dynamics are in overall very good agreement.

Performing a simulation by selecting a microcanonical ensemble at the CoI for a minimum energy crossing point between two coupled electronic states is clearly a model for the nonadiabatic dynamics. The upper electronic state is initially nonrandomly vibrationally excited by photon absorption and preparing a microcanonical ensemble at the minimum energy crossing point assumes that the intramolecular vibrational dynamics for the excited electronic state has a sufficiently long lifetime so that the nonadiabatic dynamics becomes localized at the minimum energy crossing point and a microcanonical ensemble is formed at this point. A strength of this model is that the quantum normal modes are quantized at the beginning of the simulation. In contrast, starting the trajectory calculation for the initial vibrational state prepared by photon absorption may result in unphysical zero-point energy flow⁴¹ before the nonadiabatic dynamics occurs. In future work it will be of interest to compare these two methods for initializing a nonadiabatic chemical dynamics simulation and also compare with experimental results.

■ APPENDIX: ANALYTIC EXPRESSION FOR THE CLASSICAL DENSITY OF STATES OF THE CONE AT A CONICAL INTERSECTION

The cone consists of two degrees of freedom and its classical sum of states $N(E_2)$ is given by the cone's four-dimensional phase

space volume divided by the square of Planck's constant;⁴² i.e., $N(E_2) = \text{Vol}(E_2)/h^2$. The density of states is then $\rho(E_2) = [\text{dVol}(E_2)/\text{d}E_2]/h^2$, where $\text{dVol}(E_2)/\text{d}E_2$ is the surface area of the cone's phase space volume. In the following an analytic expression for $\rho(E_2)$ is derived.

The potential energy for the upper surface of the cone is given by

$$V(x, y) = s_x x + s_y y + \sqrt{(gx)^2 + (hy)^2} \quad (\text{A1})$$

For a given value of V , this is the implicit equation of an ellipse, i.e.,

$$Ax^2 + By^2 + Cxy + Dx + Ey + F = 0 \quad (\text{A2})$$

where

$$A = g^2 - s_x^2, \quad B = h^2 - s_y^2, \quad C = -2s_x s_y, \quad D = 2s_x V, \\ E = 2s_y V, \quad F = -V^2$$

The center of the ellipse is at

$$x_c = \frac{CE - 2BD}{4AB - C^2}, \quad y_c = \frac{CD - 2AE}{4AB - C^2} \quad (\text{A3})$$

and from the expressions for A, B, C, D, E , and F it is seen that x_c and y_c are proportional to V . The tilt angle (deviation of the a -axis with respect to the x axis) is

$$\vartheta = \frac{1}{2} \arctan\left(\frac{C}{A - B}\right) \quad (\text{A4})$$

and does not depend on V . The half axes are

$$a, b = 2^{1/2} \alpha^{1/2} \left[A + B \pm \frac{C}{\sin(2\vartheta)} \right] \\ = 2^{1/2} \alpha^{1/2} \left[A + B \pm \frac{A - B}{\cos(2\vartheta)} \right] \quad (\text{A5})$$

where the $+$ and $-$ signs hold for a and b , respectively, and $\alpha = Ax_c^2 + By_c^2 + Dx_c y_c - F$.

Since the ellipse is the locus of points with potential energy equal to V , the size of the coordinate space shell with potential energies between V and $V + \text{d}V$ is $\text{d}\sigma = (\text{d}\sigma/\text{d}V)\text{d}V$, where σ is the surface of the ellipse, $\sigma = \pi\alpha(V)b(V)$. On the basis of the above formulas one obtains

$$\sigma = \pi \frac{BD^2 + AE^2 - CDE - (4AB - C^2)F}{(4AB - C^2)^{3/2}} \\ = \frac{\pi V^2}{gh(1 - g^{-2}s_x^2 - h^{-2}s_y^2)^{3/2}} = \sigma_0 V^2 \quad (\text{A6})$$

where

$$\sigma_0 = \frac{\pi}{gh(1 - g^{-2}s_x^2 - h^{-2}s_y^2)^{3/2}} \quad \text{and} \quad \text{d}\sigma = 2\sigma_0 V \text{d}V$$

The kinetic energy is $T = p_x^2/2\mu_x + p_y^2/2\mu_y$, where μ_x and μ_y are the reduced masses. (In mass weighted coordinates μ_x and μ_y are unity.) The locus of points of the momentum space with kinetic energy T is the ellipse with half axes $(2\mu_x T)^{1/2}$ and $(2\mu_y T)^{1/2}$, and the momentum space shell for an increment $\text{d}T$ is $2\pi(\mu_x \mu_y)^{1/2} \text{d}T$. The phase space shell for the increments $\text{d}V$ and $\text{d}T$ is the product $4\pi\sigma_0(\mu_x \mu_y)^{1/2} V \text{d}V \text{d}T$. Both V and T range from 0 to the energy E_2 of

the branching space, with the constraint $T + V = E_2$, so the total phase space density is the convolution of dV and dT at constant energy E_2 , i.e.,

$$\begin{aligned}\rho(E_2) &= 4\pi\sigma_0\sqrt{\mu_x\mu_y}\int\int V\delta(T+V-E_2)dVdT \\ &= 2\pi\sigma_0\sqrt{\mu_x\mu_y}E_2^2 \\ &= \frac{2\pi^2\sqrt{\mu_x\mu_y}}{gh(1-g^{-2}s_x^2-h^{-2}s_y^2)^{3/2}}E_2^2\end{aligned}\quad (\text{A7})$$

This expression does not include the division by Planck's constant squared, which is left out to avoid confusion with the standard potential energy parameter for the cone h . The sum of states for the cone, $N(E_2)$, is simply the integral of $\rho(E_2)$ from zero to E_2 and is proportional to E_2^3 .

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